

Question Bank: UNITS AND MEASUREMENT

Section 1: Fill in the Blanks / Short Answer (Conceptual)

1. The process of comparing a physical quantity with a certain basic, arbitrarily chosen, internationally accepted reference standard is called ____.

Solution: measurement

2. The units for the fundamental or base quantities are called ____ units.

Solution: fundamental (or base)

3. The SI unit of electric current is ____.

Solution: ampere (A)

4. The supplementary unit for solid angle is ____.

Solution: steradian (sr)

5. Significant figures in a measurement include all reliably known digits plus the ____ digit.

Solution: first uncertain

6. Leading zeros in a decimal number (e.g., 0.005) are ____ significant. (Are/Are not)

Solution: are not

7. Trailing zeros in a number without a decimal point (e.g., 1200 kg) are ____ significant. (Are/Are not)

Solution: are not

8. The principle stating that only physical quantities with the same dimensions can be added or subtracted is called the principle of ____ of dimensions.

Solution: homogeneity

9. The dimensional formula for acceleration is ____.

Solution: $[M L T^{-2}]$

10. The dimensional formula for energy is ____.

Solution: $[M L^2 T^{-2}]$

Section 2: Short Answer / Conceptual Questions

11. Why is the SI system of units preferred over other systems like CGS or FPS?

Solution: The SI system is preferred primarily because it is a decimal system, which makes conversions between different magnitudes of the same unit (e.g., kilograms to grams) very simple and convenient. It is also an internationally accepted standard, promoting uniformity in scientific and technological communication worldwide.

12. What are the two supplementary units in the SI system, and what physical quantities do they measure?

Solution:

- **Radian (rad):** Measures plane angles. It is defined as the angle subtended at the center of a circle by an arc equal in length to the radius.
- **Steradian (sr):** Measures solid angles. It is defined as the solid angle subtended at the center of a sphere by a surface area equal to the square of the radius of the sphere.

13. Explain the difference between accuracy and precision in measurement.

Solution:

- **Accuracy** refers to how close a measured value is to the true or accepted value of the quantity being measured. A measurement is accurate if it has a small systematic error.
- **Precision** refers to how close multiple measurements of the same quantity are to each other, indicating the reproducibility of the measurements. It also relates to the resolution of the measuring instrument (its least count). A measurement is precise if it has a small random error.
- It is possible for measurements to be precise but not accurate, or accurate but not precise, or both.

14. State the rule for determining significant figures in a number that contains trailing zeros and has a decimal point (e.g., 3.500 m).

Solution: For a number with a decimal point, all trailing zeros (zeros at the end of the number) are considered significant. This indicates that these zeros were measured and are part of the precision of the measurement. For example, 3.500 m has four significant figures.

15. What is the rule for rounding off digits if the digit to be dropped is exactly 5 and the preceding digit is even? Provide an example.

Solution: If the digit to be dropped is exactly 5, and the preceding digit is even, the preceding digit is left unchanged. The 5 is simply dropped.

- **Example:** If 2.745 is to be rounded to three significant figures, the digit 5 is dropped, and since the preceding digit (4) is even, it remains unchanged. The rounded number is 2.74.

16. Can dimensional analysis prove that a physical equation is correct? Explain why or why not.

Solution: No, dimensional analysis cannot *prove* that a physical equation is correct. It can only check for dimensional consistency. If an equation is dimensionally incorrect (the dimensions on both sides or of terms being added/subtracted are not the same), then it is definitively wrong. However, if an equation is dimensionally correct, it does not guarantee that it is physically correct, because dimensional analysis cannot account for:

1. Dimensionless constants (e.g., numerical factors like $1/2$, 2π).

2. Dimensionless functions (e.g., trigonometric, exponential, or logarithmic functions).
3. Distinguishing between physical quantities that have the same dimensions (e.g., work and torque).

17. Explain why the terms "large" or "small" are meaningless for a dimensional quantity without a specified standard for comparison.

Solution: The terms "large" or "small" are relative and subjective. A dimensional quantity (like length, mass, or speed) can only be considered "large" or "small" in comparison to another quantity of the same dimension or a defined standard. For example, a house is large compared to a car but small compared to a mountain. Without a reference point for comparison, stating that something is "large" or "small" provides no meaningful information about its actual magnitude or scale.

18. What is a "dimensional formula"? Give an example for pressure.

Solution: A dimensional formula is an expression that shows how and which of the base quantities (mass, length, time, etc.) represent the dimensions of a given physical quantity. It is typically written using the dimensional symbols (M, L, T, etc.) raised to certain powers.

- **Example for Pressure:** Pressure = Force / Area.
 - Dimensions of Force = $[MLT^{-2}]$
 - Dimensions of Area = $[L^2]$
 - Therefore, the dimensional formula for Pressure is $[MLT^{-2}]/[L^2]=[ML^{-1}T^{-2}]$.

19. Why is a set of 100 measurements of a quantity expected to yield a more reliable estimate than a set of 5 measurements only?

Solution: A larger number of measurements (like 100 compared to 5) leads to a more reliable estimate primarily because it helps to reduce the impact of **random errors**. Random errors are unavoidable variations in measurements that occur due to unpredictable fluctuations. When many measurements are taken, these random errors tend to cancel each other out when calculating the mean, leading to an average value that is statistically closer to the true value. The uncertainty in the mean value decreases as the number of measurements increases (specifically, it's inversely proportional to the square root of the number of measurements). A larger dataset also makes it easier to identify and discard any significant outliers or gross errors.

20. Describe the method of coiling a thread around a cylindrical object to estimate its diameter using a meter scale.

Solution:

1. Take a long thread and tightly coil it around a cylindrical object (like a pencil, pen, or even a meter scale itself), ensuring that the coils are touching each other without any gaps or overlaps.
2. Count the exact number of coils (N). It is recommended to make a large number of coils (e.g., 20 or 50) to minimize the percentage error.

3. Using the meter scale, measure the total length (L) occupied by these N coils.
4. The diameter (d) of the thread can then be estimated by dividing the total length (L) by the number of coils (N): $d=L/N$.

Section 3: Numerical Problems / Long Answer Questions

21. The volume of a cube of side 2 cm is equal to how many m^3 ?

Solution:

- Given side = 2 cm
- Volume = $(2 \text{ cm})^3 = 8 \text{ cm}^3$
- To convert to m^3 : $1 \text{ cm} = 10^{-2} \text{ m}$
- So, $1 \text{ cm}^3 = (10^{-2} \text{ m})^3 = 10^{-6} \text{ m}^3$
- Therefore, $8 \text{ cm}^3 = 8 \times 10^{-6} \text{ m}^3$.
- **Answer:** $8 \times 10^{-6} \text{ m}^3$

22. A vehicle moves with a speed of 36 km h^{-1} . How much distance does it cover in 2 seconds?

Solution:

- Given speed (v) = 36 km h^{-1}
- Convert speed to m s^{-1} : $36 \text{ km h}^{-1} = 36 \times \frac{1000 \text{ m}}{3600 \text{ s}} = 10 \text{ m s}^{-1}$
- Given time (t) = 2 s
- Distance covered (d) = speed $\times$ time = $10 \text{ m s}^{-1} \times 2 \text{ s} = 20 \text{ m}$
- **Answer:** 20 m

23. State the number of significant figures in the following measurements: (a) 0.00350 kg (b) $1.20 \times 10^5 \text{ m}$ (c) 5000 N (d) 20.080 s

Solution: (a) **0.00350 kg**: 3 significant figures (leading zeros are not significant, trailing zero after decimal is significant). (b) **$1.20 \times 10^5 \text{ m}$** : 3 significant figures (all digits in the mantissa are significant in scientific notation). (c) **5000 N**: 1 significant figure (trailing zeros without a decimal point are not significant). (d) **20.080 s**: 5 significant figures (zeros between non-zero digits are significant; trailing zero after decimal is significant).

24. The mass of a box measured by a grocer's balance is 3.5 kg. Two packets are added to the box with masses 120.5 g and 5.23 g. What is the total mass of the box, rounded to the correct number of decimal places?

Solution:

- Mass of box (m_{box}) = 3.5 kg

- Mass of packet 1 (m_1) = 120.5 g = 0.1205 kg
- Mass of packet 2 (m_2) = 5.23 g = 0.00523 kg
- Total mass = $m_{\text{box}} + m_1 + m_2 = 3.5 \text{ kg} + 0.1205 \text{ kg} + 0.00523 \text{ kg} = 3.62573 \text{ kg}$
- For addition, the result should have the same number of decimal places as the number with the least decimal places. m_{box} (3.5 kg) has one decimal place.
- Rounding 3.62573 kg to one decimal place gives 3.6 kg.
- **Answer:** 3.6 kg

25. The length and width of a rectangular plate are measured as 2.45 m and 1.2 m, respectively. Calculate the area of the plate to the correct number of significant figures.

Solution:

- Length (L) = 2.45 m (3 significant figures)
- Width (W) = 1.2 m (2 significant figures)
- Area (A) = $L \times W = 2.45 \text{ m} \times 1.2 \text{ m} = 2.94 \text{ m}^2$
- For multiplication, the result must be rounded to the same number of significant figures as the measurement with the least significant figures. Here, 1.2 m has 2 significant figures.
- Rounding 2.94 m^2 to 2 significant figures gives 2.9 m^2 .
- **Answer:** 2.9 m^2

26. A boy recalls the formula for the time period of a simple pendulum as $T = kl/g$. He forgets where to put the mass 'm'. Using dimensional analysis, determine if 'm' should be included in the formula.

Solution:

- Let's assume the time period T depends on length (l), acceleration due to gravity (g), and mass (m).
- Let $T \propto l^a g^b m^c \Rightarrow T = K l^a g^b m^c$, where K is a dimensionless constant.
- Write the dimensional equation:
 - $[T] = [L]^a [LT^{-2}]^b [M]^c$
 - $[M^0 L^1 T^1] = [L^a L^{b-2} T^{-2b} M^c]$
- Equating powers of dimensions on both sides:
 - For M : $c = 0$
 - For L : $a + b = 0$

- For T: $-2b=1 \Rightarrow b=-1/2$
- Substitute b into the L equation: $a+(-1/2)=0 \Rightarrow a=1/2$
- Since $c=0$, the mass 'm' does not have any power in the final formula, meaning the time period of a simple pendulum is independent of its mass.
- **Conclusion:** The mass 'm' should not be included in the formula for the time period of a simple pendulum.

27. The speed of sound in air is approximately 343 m/s. The speed of light in vacuum is approximately 3×10^8 m/s. Explain why the statement "the speed of sound is much smaller than the speed of light" is meaningful.

Solution: The statement is meaningful because it explicitly provides a standard for comparison: the speed of light. It quantifies the difference between two quantities of the same dimension (speed). When a comparison is made (e.g., "much smaller than" relative to the speed of light), it gives context to the magnitude of the speed of sound, making the statement informative and not arbitrary.

28. One mole of an ideal gas at standard temperature and pressure (STP) occupies 22.4 L (molar volume). The size (diameter) of a hydrogen molecule is about 1 Å. Calculate the ratio of the molar volume to the atomic volume of a mole of hydrogen molecules. Explain why this ratio is so large. (Given: 1 L = 10^{-3} m³, 1 Å = 10^{-10} m, Avogadro's number $N_A = 6.022 \times 10^{23}$ molecules/mole)

Solution:

- **Molar volume of gas ($V_{\text{molar,gas}}$):** 22.4 L = 22.4×10^{-3} m³
- **Atomic volume of a mole of hydrogen molecules ($V_{\text{atomic,mole}}$):**
 - Diameter of hydrogen molecule (d) ≈ 1 Å
 - Radius (r) = $d/2 = 0.5$ Å = 0.5×10^{-10} m
 - Volume of a single hydrogen molecule (V_{molecule}) (assuming spherical) = $(4/3)\pi r^3$
 $V_{\text{molecule}} = (4/3) \times 3.14159 \times (0.5 \times 10^{-10} \text{ m})^3 \approx 0.5236 \times 10^{-30} \text{ m}^3$
 - Volume of a mole of hydrogen molecules ($V_{\text{atomic,mole}}$) = $N_A \times V_{\text{molecule}}$
 $V_{\text{atomic,mole}} = (6.022 \times 10^{23}) \times (0.5236 \times 10^{-30} \text{ m}^3) \approx 3.153 \times 10^{-7} \text{ m}^3$
- **Ratio:** Ratio = $V_{\text{molar,gas}}/V_{\text{atomic,mole}} = (22.4 \times 10^{-3} \text{ m}^3)/(3.153 \times 10^{-7} \text{ m}^3)$ Ratio $\approx 7.104 \times 10^4 \approx 7.1 \times 10^4$
- **Explanation for large ratio:** This ratio is very large because, in a gas at STP, the actual volume occupied by the molecules themselves is negligible compared to the total volume the gas occupies. The vast majority of the molar volume of a gas is empty space between the molecules. Gas molecules are widely dispersed and move freely, leading to a much larger overall volume compared to the actual volume of the constituent particles.

29. The Sun is a hot plasma with a mass of 2.0×10^{30} kg and a radius of 7.0×10^8 m. Calculate its average mass density. Based on the calculated density, would you classify the Sun's matter as being in the density range of solids, liquids, or gases? Explain your reasoning.

Solution:

- **Mass of the Sun (M):** 2.0×10^{30} kg
- **Radius of the Sun (R):** 7.0×10^8 m
- **Volume of the Sun (V):** Assuming spherical, $V = \frac{4}{3}\pi R^3$ $V = \frac{4}{3} \times 3.14159 \times (7.0 \times 10^8 \text{ m})^3$
 $V = \frac{4}{3} \times 3.14159 \times (343 \times 10^{24}) \text{ m}^3 \approx 1.437 \times 10^{27} \text{ m}^3$
- **Average Mass Density (ρ):** $\rho = M/V$ $\rho = (2.0 \times 10^{30} \text{ kg}) / (1.437 \times 10^{27} \text{ m}^3) \approx 1.3917 \times 10^3 \text{ kg m}^{-3}$
- Rounding to 2 significant figures (based on input values of 2.0 and 7.0): $\rho \approx 1.4 \times 10^3 \text{ kg m}^{-3}$ or 1400 kg m^{-3} .
- **Classification:**
 - Typical gas densities are around 0.1 kg m^{-3} to 2 kg m^{-3} .
 - Typical liquid densities are around 800 kg m^{-3} to 1500 kg m^{-3} (e.g., water is 1000 kg m^{-3}).
 - Typical solid densities are higher, from 1000 kg m^{-3} upwards.
- The calculated density of approximately 1400 kg m^{-3} falls within the range of **liquids** and some less dense solids.
- **Reasoning:** Despite the extremely high temperatures (which would normally lead to low gas densities), the Sun's enormous gravitational force compresses its plasma intensely. This immense pressure forces the particles much closer together than in a typical gas, resulting in a density comparable to that of liquids and even some solids on Earth. The matter is in a plasma state, but it is an extremely dense plasma.

30. The displacement of a particle is given by the equation $x = Pt + Qt^2 + Rt^3$, where x is displacement in meters and t is time in seconds. Use dimensional analysis to find the dimensions of the constants P , Q , and R .

Solution:

- According to the principle of homogeneity of dimensions, each term in the equation must have the same dimensions as the quantity on the left side, which is displacement $[x]$.
- Dimensions of displacement $[x] = [L]$.
- 1. **For the term Pt :**
 - Dimensions of $Pt = [L]$
 - $[P][T] = [L]$

- $[P]=[L]/[T]=[LT^{-1}]$
- **Dimensions of P are $[LT^{-1}]$ (which are the dimensions of velocity).**

2. **For the term Qt^2 :**

- Dimensions of $Qt^2=[L]$
- $[Q][T^2]=[L]$
- $[Q]=[L]/[T^2]=[LT^{-2}]$
- **Dimensions of Q are $[LT^{-2}]$ (which are the dimensions of acceleration).**

3. **For the term Rt^3 :**

- Dimensions of $Rt^3=[L]$
- $[R][T^3]=[L]$
- $[R]=[L]/[T^3]=[LT^{-3}]$
- **Dimensions of R are $[LT^{-3}]$ (which are the dimensions of jerk, or rate of change of acceleration).**